

Unified Military Analytics and Comparison Dashboard-DV

Report for Milestone 3

**INTERNSHIP OF
DATA VISUALIZATION
IN
INFOSYS SPRINGBOARD**

Submitted by
Bhagya P Bijay Kumar

Table of Content

Sr no.	Title	Page No.
1	Abstract	3
2	Module 5: 5.1 Quick Stats 5.2 Nation Overview	4-7
3	Module 6: 6.1 Compare Powers 6.2 Coalition Builder	8-11
4	Conclusion	11

ABSTRACT

This project aims to develop a fully interactive dashboard suite for analyzing global military power in 2025. Using Python, military metrics will be scraped from GlobalFirepower.com for 140+ countries via a pre-defined list (links_for_military_data.txt).

The project leverages Python-based data pipelines to scrape, clean, and process open-source defense data from GlobalFirepower.com. Unlike traditional single-platform dashboards, GDM emphasizes cross-platform flexibility — allowing deployment in tools like Tableau, Power BI, or Python-based web apps built using Streamlit or Dash.

Dashboards planned:

- Quick Stats – A global overview of rankings, trends, and highlights.
- Nation Overview – Detailed analysis of individual country capabilities.
- Compare Powers – Side-by-side military and economic comparison.
- Coalition Builder – Interactive simulation of alliance strength and combined assets.

The final deliverable is a comprehensive analytics suite that enables users to compare nations, evaluate coalitions, and analyze KPIs like Power Index Rank Gap, Assets per Capita, and Defense Budget-to-GDP Ratio through intuitive and dynamic visualizations.

Milestone 3: Full Dashboard Development

Module 5: Build Quick Stats and Nation Overview Dashboards

5.1 Quick Stats

Steps:

- Load the final KPI dataset.
- Identify and verify Measures and Dimensions.
- Identify and list all KPIs required for the Quick Stats dashboard.
- Filters:
 - Region
 - Continent
 - Alliance
- Cards:
 - Total Country
 - Total Defense Budget
 - Total Military Manpower
 - Budget to GDP Ratio
 - Assets per capita
- Map:
 - Top 5 Countries by Total military manpower
 - Dimension: country
 - Measure: sum of military manpower
- Donut Chart:
 - Top 5 Powers by Active Personnel
 - Dimension: country
 - Measure: sum of Active Personnel
- Pie Chart:
 - Top 5 Countries by defense budget
 - Dimension: country
 - Measure: sum of defense budget
- - Top 5 Countries by assets per capita
 - Dimension: country
 - Measure: sum of assets per capita

- Area Chart:
 - Top 10 countries by power index
 - Dimension: country
 - Measure: sum of power index

Use filter in everything for example:

The screenshot shows a filter configuration panel for the dimension 'country'. The filter is set to 'top 5 by Sum of activ...'. The 'Filter type' is set to 'Top N'. The 'Show items' section shows 'Top' and '5'. The 'By value' section shows 'Sum of active_person...'. An 'Apply filter' button is at the bottom right.

Figure: Top N filters

So in quick stats we can filter on the basis of nato alliance, continent and region and visualize the top countries on the basis of various metrics.

QUICK STATS DASHBOARD

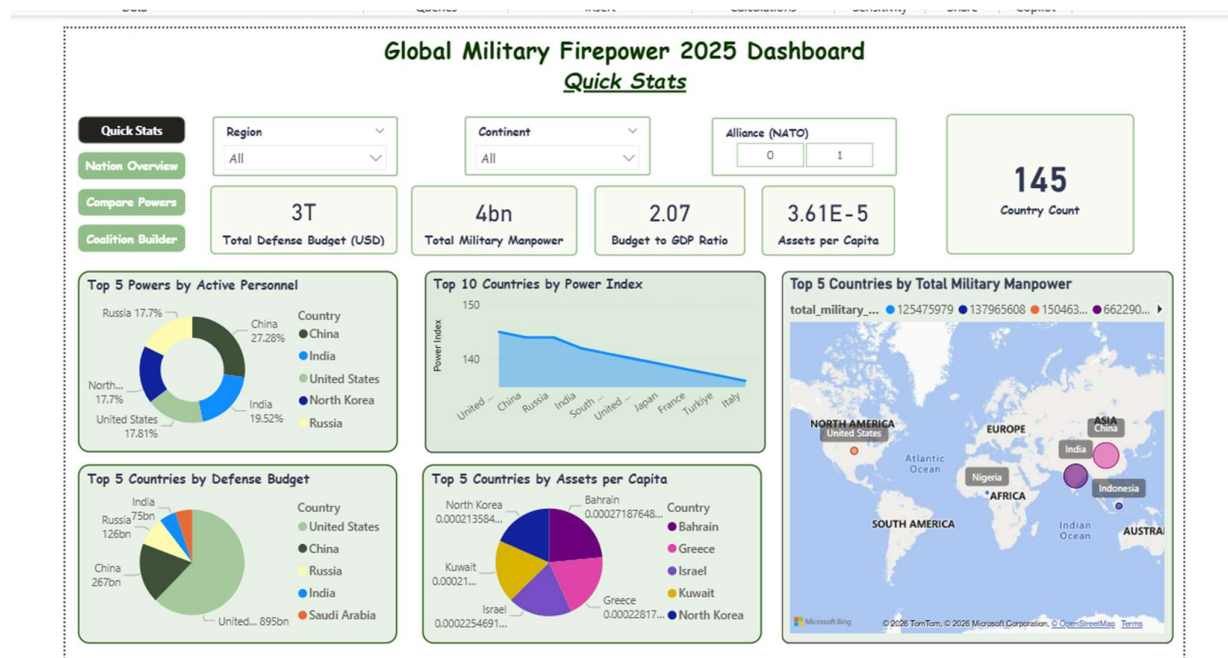


Figure: Quick Stats Dashboard

5.2 Nation Overview

Steps:

- Identify and list all KPIs required for the Nation Overview dashboard.
- Filters:
 - Country
 - Alliance

So in the Nation Overview dashboard, the visualizations can be filtered with the help of country filter, to know about each country.

- Cards:
 - Total land area (sq_km)
 - Total Population (sum)
 - Country: to show the selected country
 - Budget per soldier
 - Budget to gdp ratio
 - Multi row card: country, pwr index and power rank
- Map:
 - Country and Region
 - Dimension: Country
 - Measure: Region
- Bar Chart:
 - Defense Budget
 - Dimension: country
 - Measure: defense_budget_usd, country with highest budget
 - Using modelling, new measure:

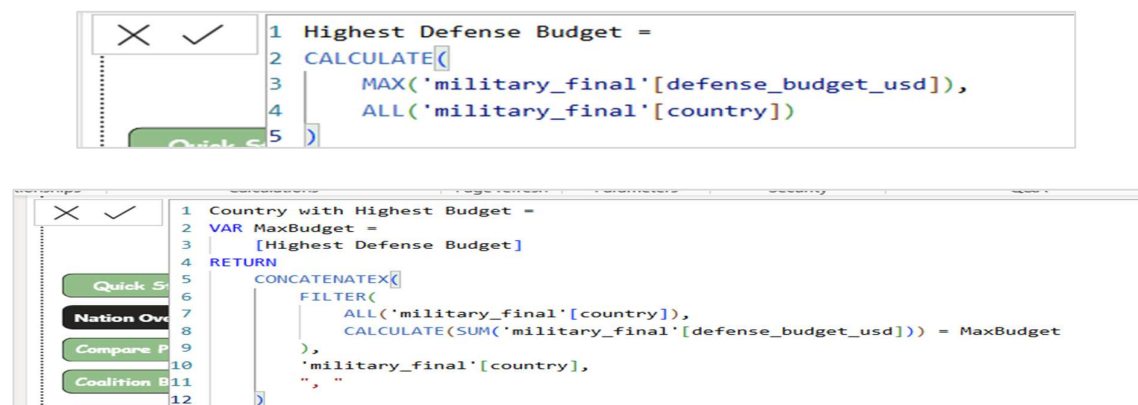


Figure: New Measure: Country with Highest Budget

- Pie Chart:
 - Active vs Reserve vs Paramilitary
 - Dimension: country
 - Measure: Paramilitary, Reserve_personnel, Active_personnel
 - Tooltips: Country and Power Rank
- Gauge:
 - Military Participation Ratio
 - Sum of military_participation_ratio

NATION OVERVIEW DASHBOARD

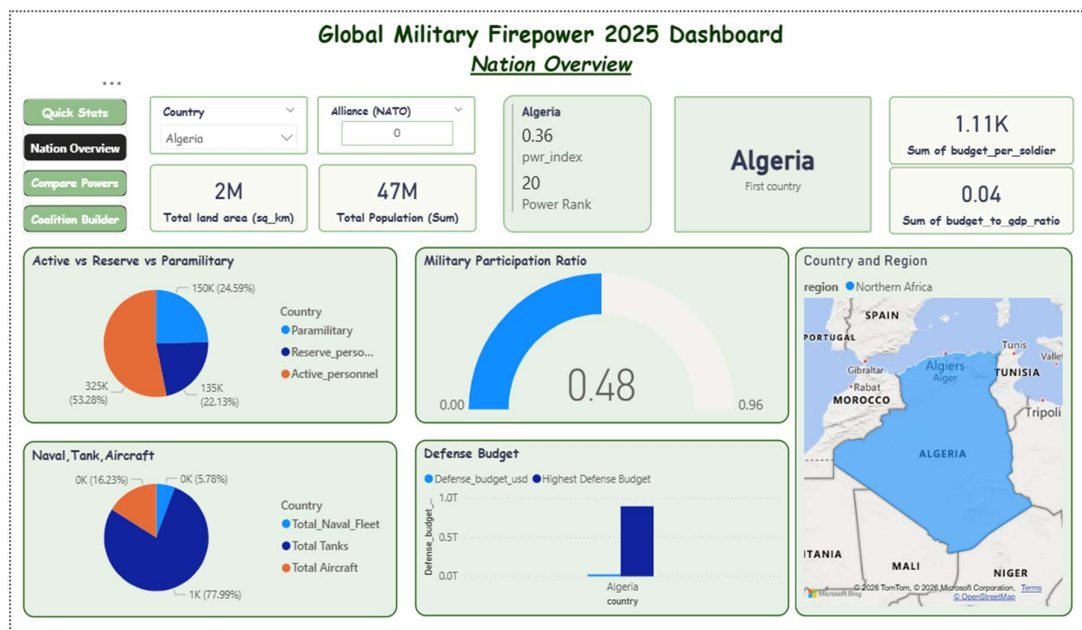


Figure: Nation Overview Dashboard

Module 6: Build Compare Powers and Coalition Builder Dashboards

6.1 Compare Powers

In this dashboard, two countries can be selected with the help of filters, and compare these countries on the basis of various power related metrics like, aircraft, tanks, naval etc.

Steps:

- Load the final KPI dataset.
- Identify and verify Measures and Dimensions.
- Identify and list all KPIs required for the Compare Powers dashboard.
- Filters: Change the settings as select one.
 - Country A
 - Country B

For the selected country to filter the correct visual, select the filter visual, go to format, there edit interactions, there filter or none was selected according to requirements.

- Cards:
 - Country A selected
 - Country B selected
 - Total Military Manpower A
 - Total Military Manpower B
 - Budget to GDP Ratio A and B cards
 - Assets per capita A and B cards
 - Power index rank gap A and B cards.
- Bar chart:
 - i. Defense Budget (USD) Country A: country and defense_budget_usd
 - ii. Defense Budget (USD) Country B: country and defense_budget_usd
 - iii. Total Aircraft, Tanks and Naval Fleet by Country A

For total_aircrafts:

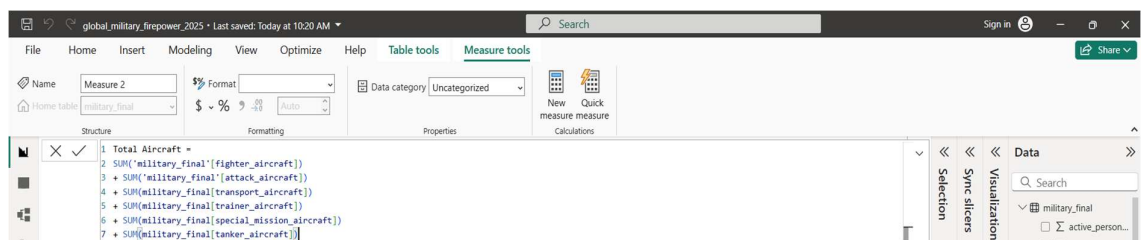


Figure: New measure: total_aircraft

- iv. Total Aircraft, Tanks and Naval Fleet by Country B
- v. Active Personnel: Country A
- vi. Active Personnel: Country B

Add data labels to all the charts.

COMPARE POWERS DASHBOARD

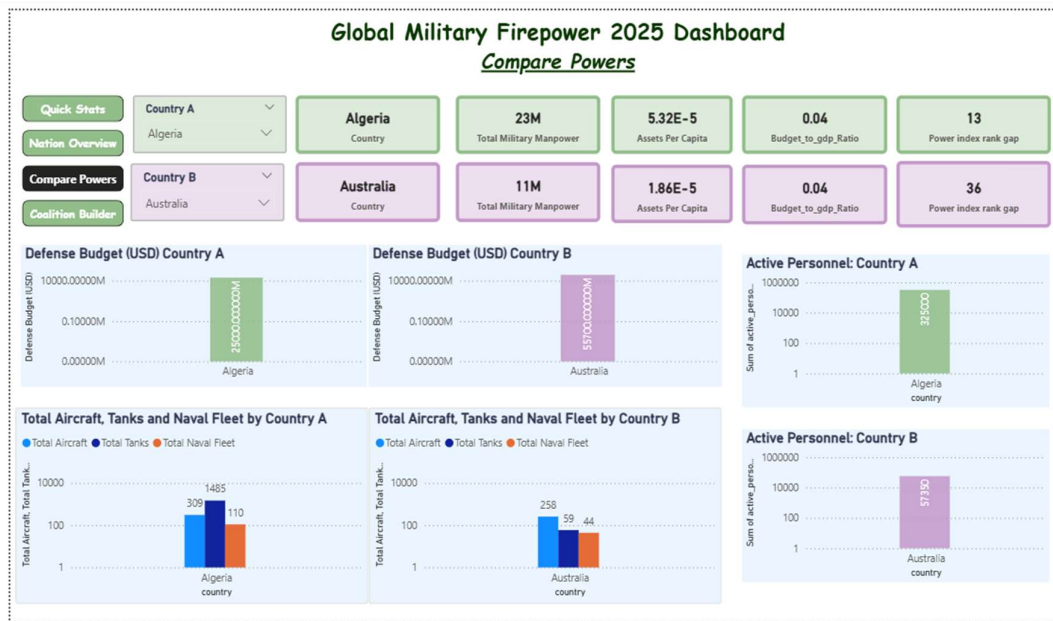


Figure: Compare Powers Dashboard

6.1 Coalition Builder

In this dashboard we can compare the power of the countries by combining two or three countries and comparing with other combination of countries, to find out which has better combined strength.

Steps:

- Load the final KPI dataset.
- Identify and verify Measures and Dimensions.
- Identify and list all KPIs required for the Coalition Builder.
- Filters: Change the settings as MULTIPLE SELECT.
 - Coalition A
 - Coalition B

For the selected country to filter the correct visual, select the filter visual, go to format, there edit interactions, there filter or none was selected according to requirements.

- Cards:
 - 2 Multiple row cards for selected countries.
 - Total Military Manpower for coalition A.
 - Total Military Manpower for coalition B.
 - Total Land area sq km for coalition A.
 - Total Land area sq km for coalition B
 - Total Assets per capita for coalition A.
 - Total Assets per capita for coalition B.
 - Total Defense Budget (USD) for coalition A.
 - Total Defense Budget (USD) for coalition B.
- Bar chart:
 - i. Total Naval Fleet (Coalition of countries A): sum_of_total_naval_fleet
 - ii. Total Naval Fleet (Coalition of countries B): sum_of_total_naval_fleet
 - iii. Total Aircraft (Coalition of countries A): sum_of_total_aircraft
 - iv. Total Aircraft (Coalition of countries B): sum_of_total_aircraft
 - v. Total Tanks (Coalition of countries A): sum_of_tanks
 - vi. Total Tanks (Coalition of countries B): sum_of_tanks

COALITION BUILDER DASHBOARD



Figure: Coalition Builder Dashboard

Conclusion:

This concludes the report for milestone 3 including building of 4 different dashboards: quick stats, nation overview, compare powers and coalition builder, with each page having various filters, visualizations, cards, and navigation buttons completing the third milestone of the data visualization internship of Infosys Springboard.